

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

Name: Pravin R

Reg. No: 192421292

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Experiment 11: Credit Score Classification using Machine Learning

Aim

To implement a machine learning program for classifying customers into different credit score categories based on their financial attributes.

Procedure

1. Import the required libraries.
2. Create a sample credit score dataset.
3. Split the dataset into training and testing sets.
4. Train a Random Forest classification model.
5. Predict the credit score category of test data.
6. Calculate the classification accuracy.
7. Classify a new customer.
8. Display the result.

Algorithm

1. Start.
2. Load the credit score dataset.
3. Separate input features and target classes.
4. Split the dataset into training and testing data.
5. Train the classification model.
6. Predict credit score classes for test data.
7. Evaluate the model using accuracy.
8. Predict the credit score category for a new customer.
9. Stop.

Code

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

X = [
    [25, 25000, 1, 2],
    [30, 40000, 3, 1],
    [35, 60000, 5, 0],
    [45, 80000, 8, 0],
    [50, 90000, 10, 0],
    [22, 20000, 0, 4],
    [28, 30000, 2, 3],
    [40, 70000, 7, 1],
    [55, 100000, 12, 0],
    [24, 22000, 1, 4],
    [32, 50000, 4, 1],
    [48, 85000, 9, 0],
    [27, 28000, 1, 3],
    [38, 65000, 6, 1],
    [52, 95000, 11, 0]
]

y = [
    "Poor", "Average", "Good", "Good", "Good",
    "Poor", "Poor", "Good", "Good", "Poor",
    "Average", "Good", "Poor", "Good", "Good"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Labels:", y_test)
print("Predicted Labels:", y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

new_customer = [[35, 55000, 5, 1]]
prediction = model.predict(new_customer)
print("Credit Score Classification:", prediction[0])
```

Output

```
*** Actual Labels: ['Poor', 'Good', 'Poor', 'Good', 'Poor']
Predicted Labels: ['Poor' 'Good' 'Poor' 'Good' 'Poor']
Accuracy: 1.0
Confusion Matrix:
[[2 0]
 [0 3]]
Credit Score Classification: Average
```

Result

The Credit Score Classification program was successfully implemented using a Random Forest classifier. The model classified customers into Poor, Average, and Good credit score categories and predicted the category of a new customer.